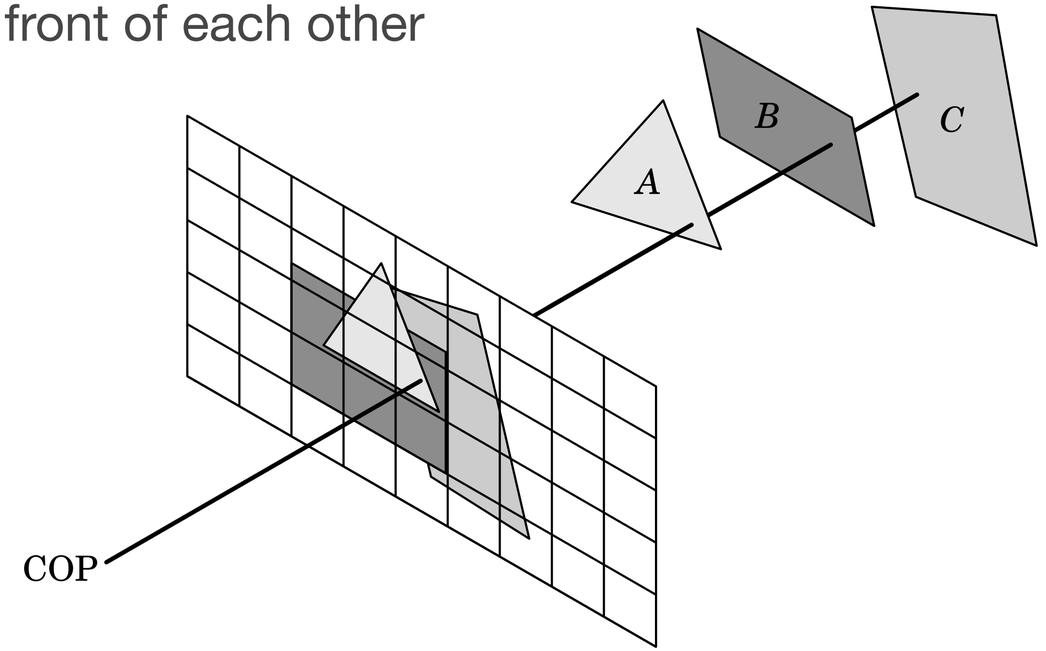
Visibility

e.g. in Assassin's Creed I, 2007

Visibility Problem

- Identify which objects or individual geometric elements, polygons, of surfaces are visible or occluded from a given viewpoint
  - fundamental problem of 3D computer graphics
- Visibility defines what is inside the view frustum in in front of each other
  - a 3D geometry intersection problem (culling)
  - a layering sorting problem
- Solving visible occlusion is a.k.a.
  - Hidden Surface Determination
  - Hidden Surface Removal
  - Visible Surface Determination



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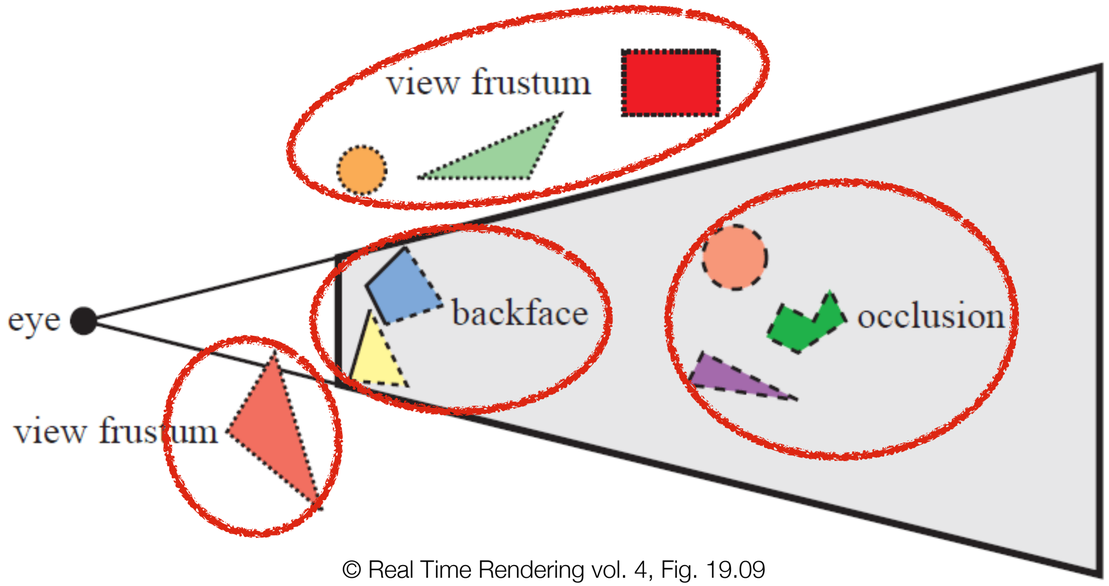
Culling



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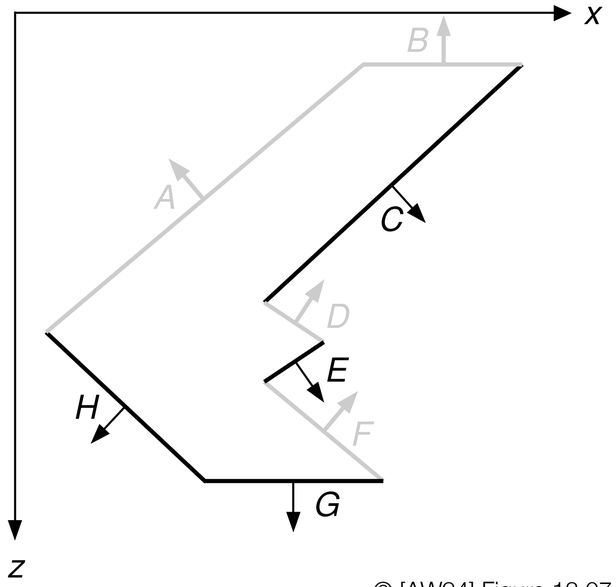
- Identify primitives which cannot be visible and can be reject the before processing and rendering
  - reduces load on the graphics subsystem
- View frustum culling (clipping)
  - reject objects or primitives outside of the view frustum
  - clip as needed if element is intersecting the view frustum
- Back-face culling
  - remove one-sided polygons if viewed from the back/ inside
- Occlusion culling
  - avoid rendering primitives that are occluded by other objects

Back-Face Culling

- Front-facing if viewer can see the visible side of a polygon
  - normal is oriented towards the viewer
- Use sign of dot-product between normal and vector to viewpoint
  - in parallel-projection canonical view volume back-facing
    simply determined by the normal's negative z-component
- Polyhedra and closed manifold surface meshes enclose a solid volume
  - all back-facing polygons are fully occluded by front-facing
  - around 50% of all faces may be culled



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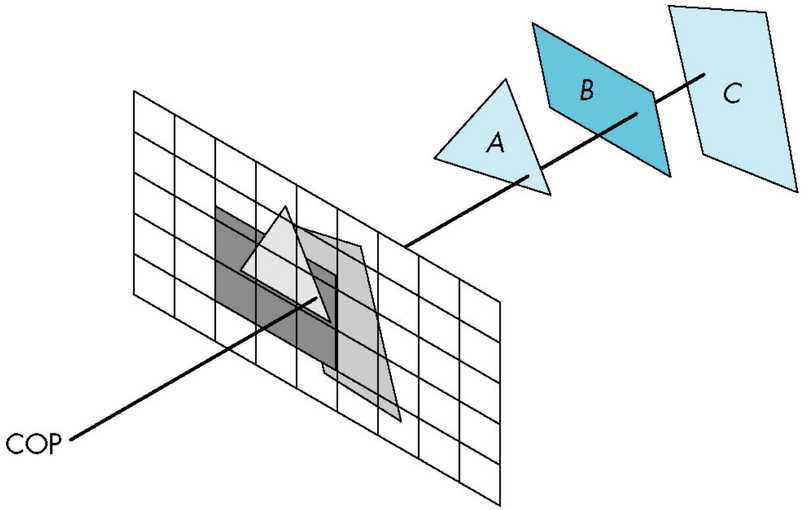
Visible Surface Determination

- VSD is eventually needed at the fragment level
  - results in a set of visible pixels for each graphics primitive
- Complex and potentially time-consuming task

Two basic approaches:

- Image-precision (image-space)
  - performed at a fixed resolution of the discrete rasterization
  - aliasing problems in visibility calculations
- Object-precision (object-space)
  - performed at the precision of the object definition
  - complex geometry processing operations



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Visible Surface Determination Approaches

1. For each pixel (image-space)
   - determine the surface that is intersected by the pixel projector (ray) that is closest and draw its color

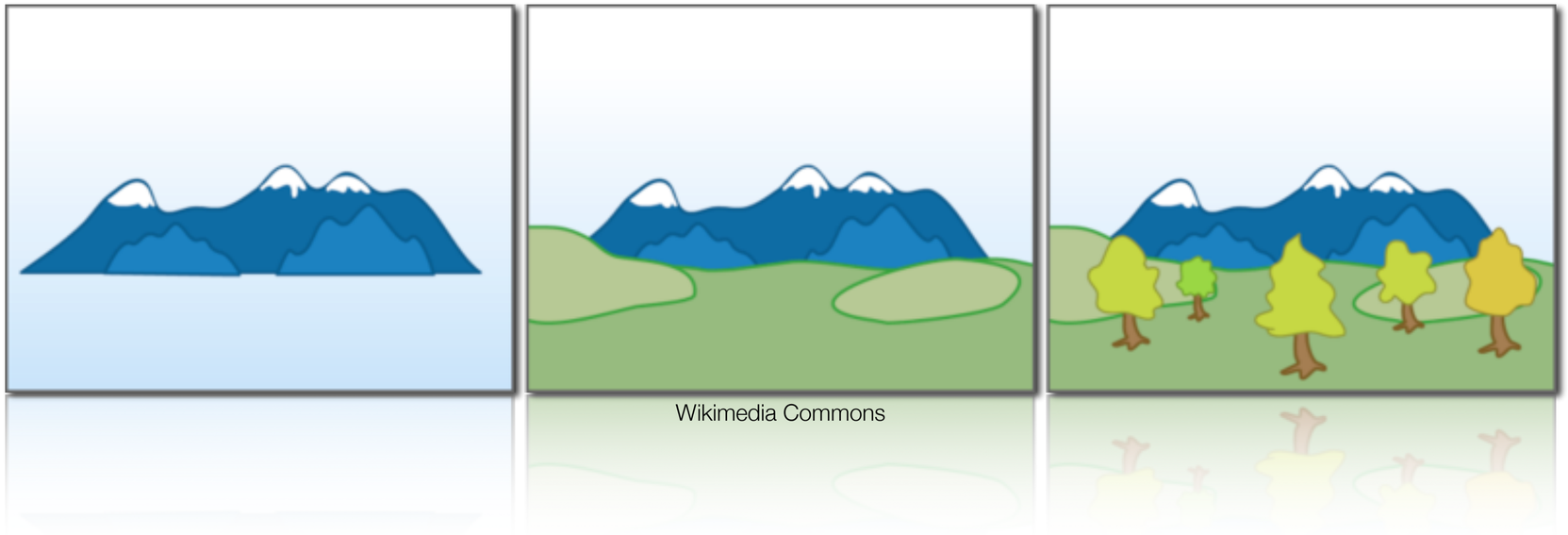
2. For every surface (object-space)
   - test against itself and all others for occlusion, determine the region that is visible and fill by color

- First approach generally needs $\mathrm{O}(np)$ time and second $\mathrm{O}(n^2)$ for n surfaces and p pixels
  - not clear which is best in general
  - even though for high-resolution displays with $n << p$ the approach may seem superior,
    its algorithmic complexity is higher → slower

Past and Now

- VSD problem has historically been addressed by clever and computationally efficient algorithms
  - ordering of geometric objects
  - recursive subdivision of object or screen space
- Modern practice relies mostly on simple and fast hardware accelerated solutions
  - fast fragment level determination of closest object
- … or raytracing
  - in contrast to the real-time rasterization pipeline
  - solves explicit ray-object intersections problems

Painter’s Algorithm



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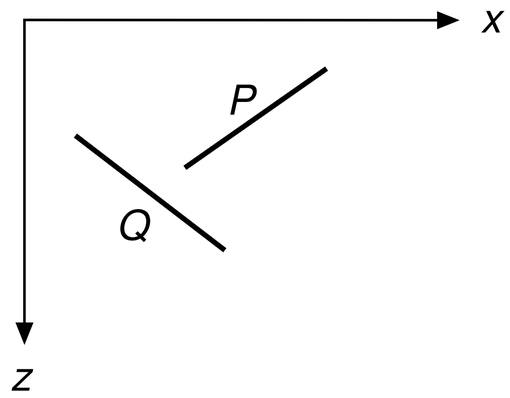
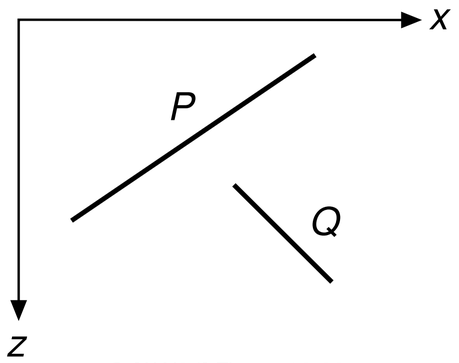
- Depth sorting of polygons followed by back-to-front drawing,
  i.e. with decreasing distance of polygon to viewpoint
  1. sort polygons (e.g. by their center of mass)
  2. resolve depth ambiguities
  3. scan convert polygons back-to-front (like painting on top of all previous ones)

Polygon Sorting Order

- Painting order is clear if polygons have no depth-interval overlap along the z-axis
  - all vertices of one polygon are clearly in front of the other polygon's vertices
- Other configurations of polygons may still have a clear ordering
- If overlapping in z, P can safely be drawn before Q upon first true test of:
  1. do the polygons not overlap in the x coordinate?
  2. do the polygons not overlap in the y coordinate?
  3. is P completely on the opposite, far side of Q from the view point?
  4. is Q entirely on the same near side of P as the view point?
  5. do the projections of the two polygons not overlap?
- Invert order for tests 3 and 4





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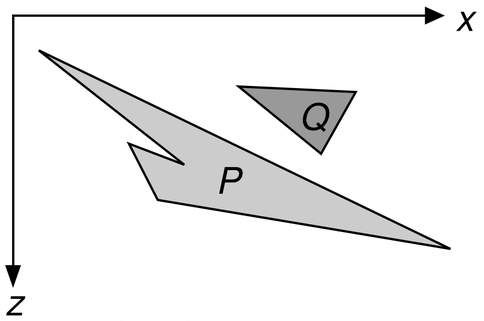

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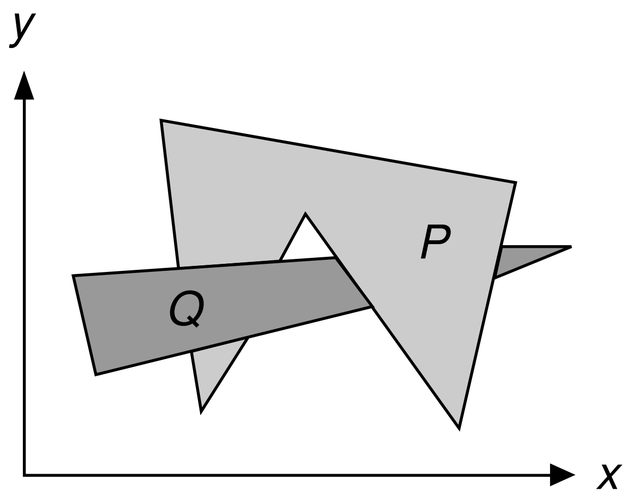
Depth Ambiguities

- If no simple ordering can be found, at least one polygon must be split into two parts
  - split by intersecting one polygon by the plane of the other

- Depth-cycles must be detected and resolved
  - mark visited polygons to detect a cycle
  - split at least one polygon to break a cycle



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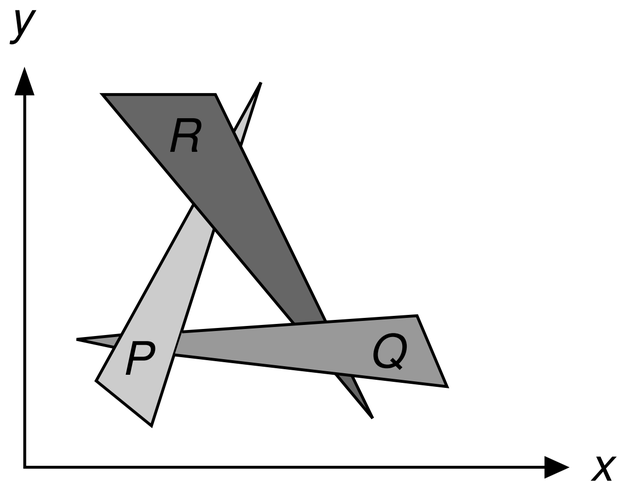

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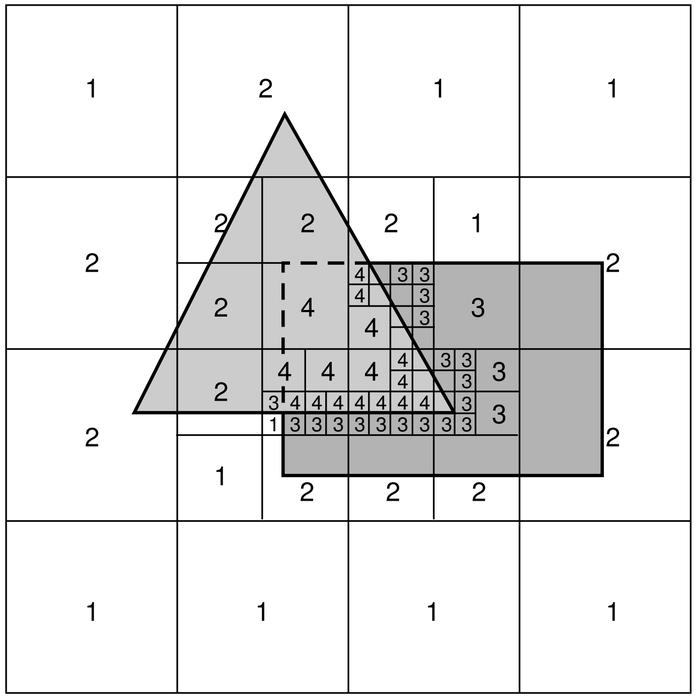
Warnock’s Algorithm

- Divide and conquer approach to the visibility problem based on spatial partitioning of the projection or image plane
  - recursively subdivide screen area into cells until a cell is easy to draw
- Quadtree subdivision into cells is performed until reaching the resolution of the raster image in the limit
  - fewer polygons overlap smaller quadtree cell regions
  - ultimately a decision on the fragment level can be reached



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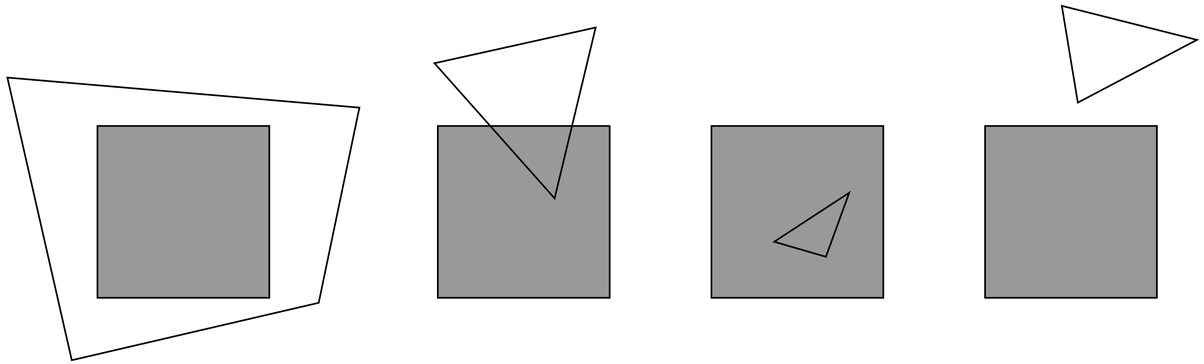
Polygon-Cell Overlap Tests



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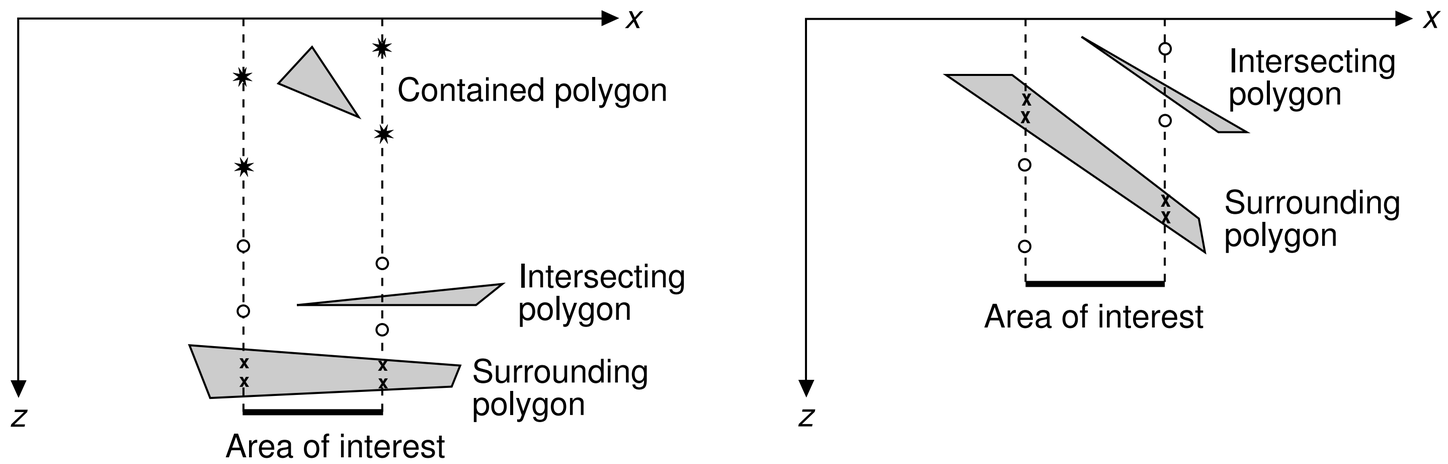
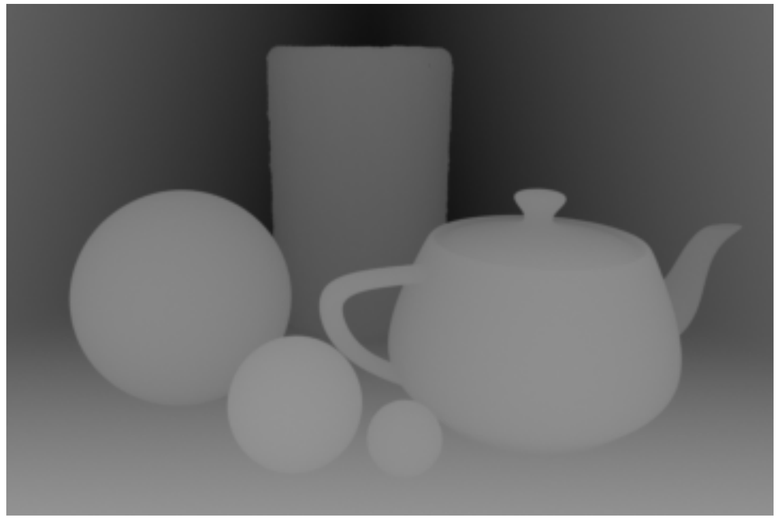
- Cell subdivision is based on 4 simple polygon-cell intersection cases
- Evaluation of four basic cell configurations:
  1. **all** polygons disjoint form the cell → leave background
  2. **one** intersecting or contained polygon → paint clipped polygon
  3. **one** single surrounding polygon → fill cell with polygon color
  4. **one** surrounding polygon in front of all others → fill cell with polygon color
- Else recursively subdivide cell and test

Surrounding Polygon Depth Test (Case 4)



- Examine the z coordinates of all polygon planes at the corners of the cell
  - determine if one infinite plane is clearly in front of the others for this specific cell only
- If there is a surrounding polygon whose four corner z values are all closer to the viewer then every other polygon, then it is in front of all of them
  - evaluate all polygon plane equations for the z-value given all four corner x,y-values of the cell
- For fragment level cell, test is performed only at the center position

z-Buffer Algorithm



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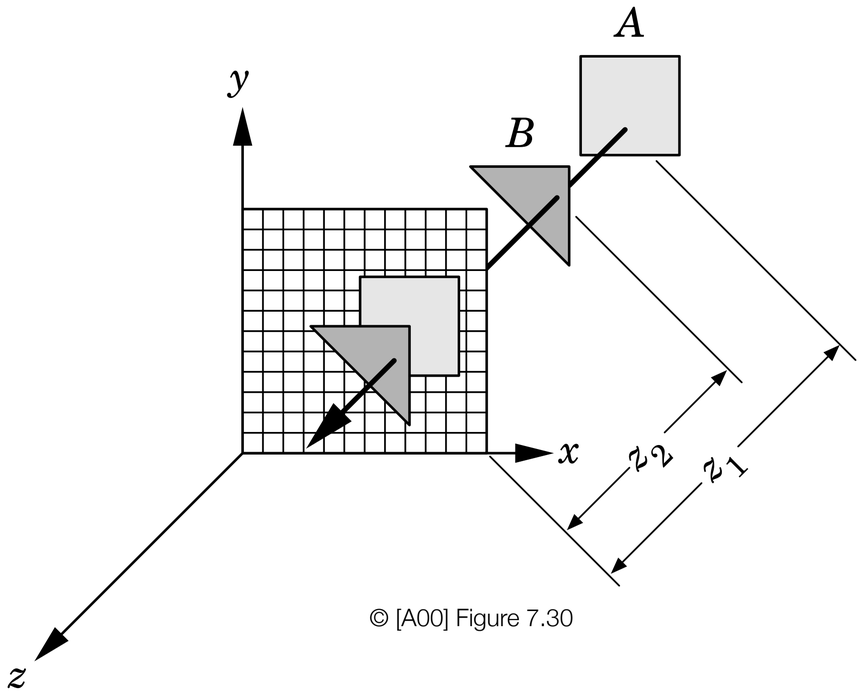

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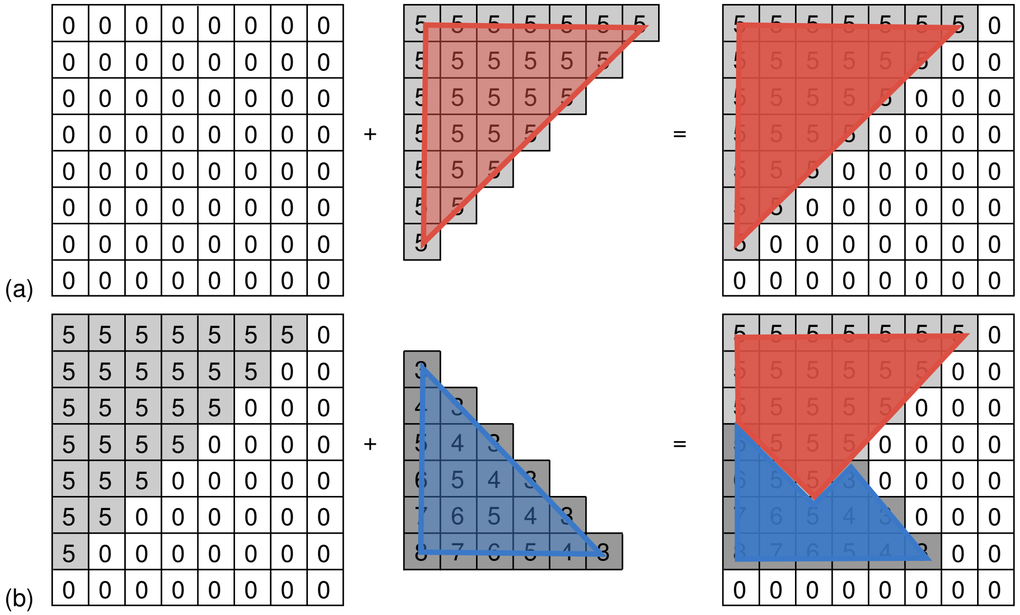
- During rasterization of each polygon, the fragment's depth value is written into the z-buffer, in addition to the color that is written into the frame buffer
  - z-buffer stores depth, the distance from the camera, in addition to color values per fragment
- Image-precision approach that determines the visibility per fragment using the values in the z-buffer
  - for each new fragment, simply compare the new fragment's z-value with the one previously stored in the z-buffer

z-Buffer Algorithm



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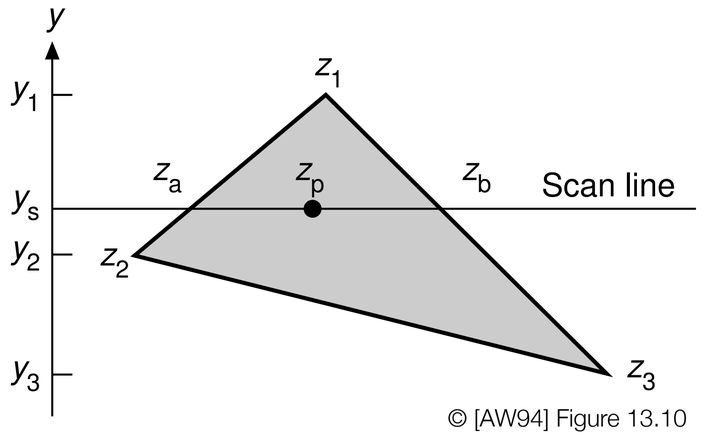
Example from Foley et al. book uses **larger values being closer to the viewer**.

- Initialize frame and depth-buffers to background color and maximal z-depth
- Foreach polygon
  - rasterize polygonal into fragments
  - foreach generated fragment
    - write fragment color and depth, if and only if the new depth is closer to the camera than the value stored in the z-buffer

Depth Interpolation

- Per-fragment depth calculation can be integrated with scan-conversion during the rasterization stage
  - exploit planarity of polygonal faces and depth-coherence on smooth surface patches
- Over a triangle, the z-coordinate can be linearly interpolated along the edges between their endpoints and across the scan-line between the two scan-line intersections
  - incremental calculation from one scan-line to the next



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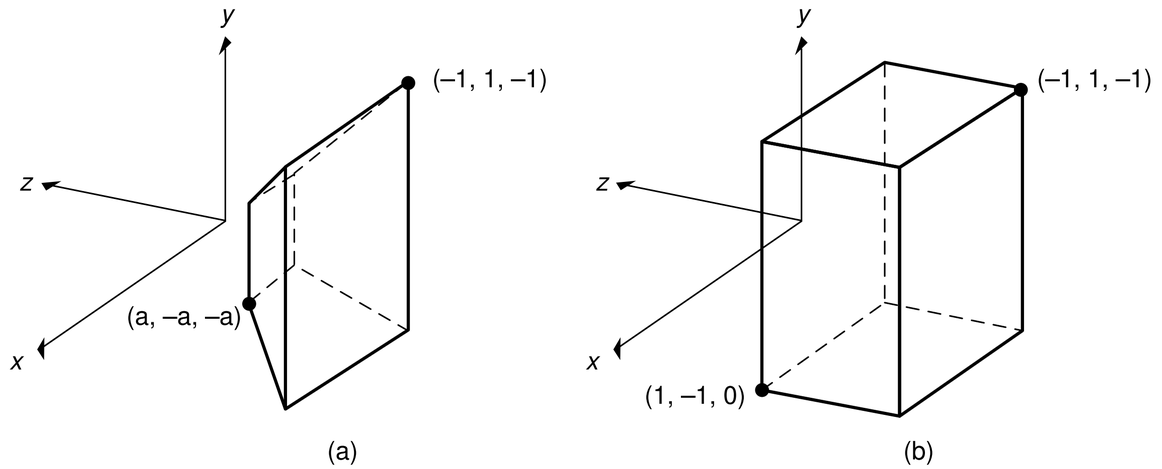
$$z_a = z_1 - (z_1 - z_2)\frac{y_1 - y_s}{y_1 - y_2}$$

$$z_b = z_1 - (z_1 - z_3)\frac{y_1 - y_s}{y_1 - y_3}$$

$$z_p = z_b - (z_b - z_a)\frac{x_b - x_p}{x_b - x_a}$$

z-Buffer Algorithm



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- Operates effectively in normalized parallel-projection canonical view volume
  - pixel corresponds to projection ray parallel to the z-axis, hence z-value directly represents depth distance
- Polygons can be rasterized in arbitrary order
  - any pixel in frame buffer will be overwritten if new fragment is closer to viewpoint than current z-buffer depth value
  - z-buffer has to be initialized to the back clipping plane

Properties

- **image precision method** with fixed display and depth resolution
  - prone to aliasing problems and flickering effects from numerical inaccuracies
  - depth resolution has to match range of possible z-values
- **no presorting** required and easy to implement
  - polygons are drawn in their processing order
  - integrated with scan conversion, depth can be computed incrementally
- **no object comparisons**
  - one depth comparison per pixel and polygon containing it
  - processing depending on fill-rate, number of generated fragments
- **no explicit intersection tests**
  - visibility can be determined on any type of surface that can be rasterized
- **buffer records smallest z-depth** found so far for a given x,y
  - search over fixed x,y coordinate to find smallest z-depth
- **color does not have to be computed if fragment is not visible**
  - complex illumination and shading calculations can be avoided for hidden fragments (if surfaces are processed front-to-back)

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- [A00] Electronic Book Support for Interactive Computer Graphics: A Top-Down Approach Using OpenGL by Edward Angel. Copyright E. Angel 2000. (http://www.cs.unm.edu/~angel/BOOK/SECOND_EDITION/)
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